ARJUNA 2.0

-JEE 2024 Exam-



- Physics
- Kinetic Theory of Gases and Thermodynamics



Lecture No.- 03

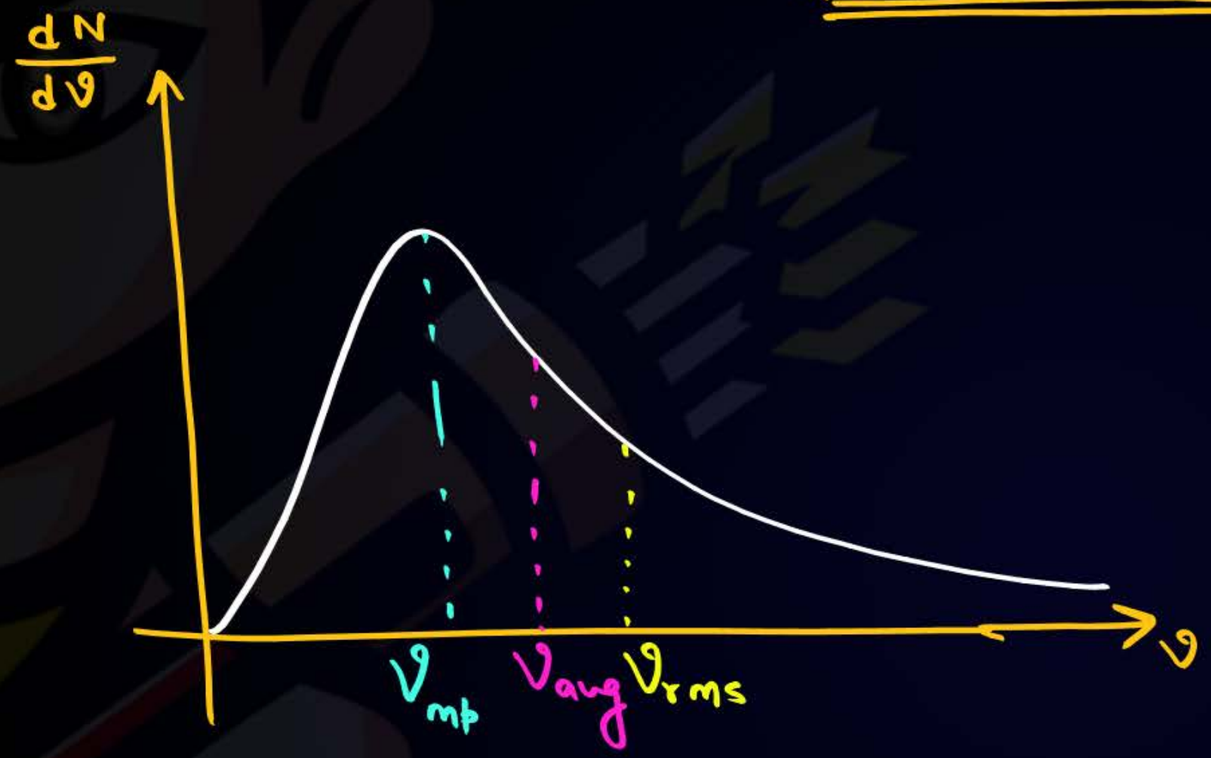
By- Himanshu Gupta



Today's Targets

1. Pressure Exerted by Gas
2. Kinetic Energy of Gas
3. Mean Free Path
4. Gas Laws

Quick Revision



$$v_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2k_BT}{m}}$$

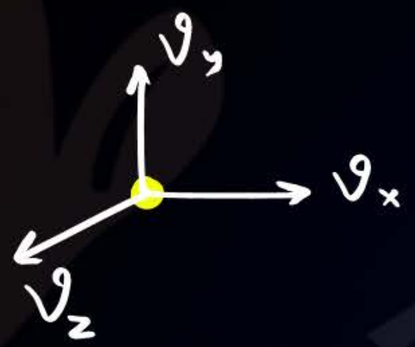
$$\left[k_B = \frac{R}{N_A} \qquad m = \frac{M}{N_A}\right]$$

$$v_{avg} = \sqrt{\frac{8RT}{\pi M}}$$

Root Mean Square speed →

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$v_{rms} = \sqrt{\frac{\Sigma v_i^2}{N}}$$



$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\Rightarrow v^2 = v_x^2 + v_y^2 + v_z^2$$

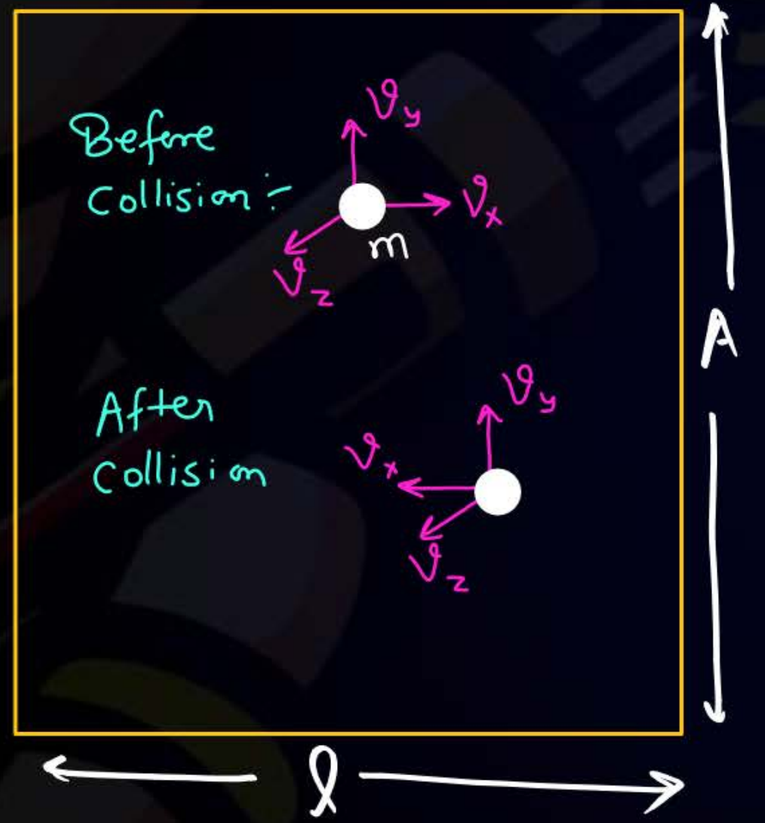
$$\Rightarrow \Sigma v^2 = \Sigma v_x^2 + \Sigma v_y^2 + \Sigma v_z^2$$

$$\left[\frac{\Sigma v^2}{N}\right] = 3\left[\frac{\Sigma v_x^2}{N}\right]$$

$$v_{rms}^2 = 3\, v_{x\text{-}rms}^2$$

Pressure Exerted by Gas

Derivation - School



$|\text{Change in momentum of one molecule}| = 2mv_x$

Time b/w two successive collisions, $T = \frac{2\ell}{v_x}$

Avg. force exerted by one molecule, $f_0 = \frac{\Delta P}{\Delta t} = \frac{2mv_x}{\left(\frac{2\ell}{v_x}\right)}$

$$f_0 = \frac{mv_x^2}{\ell}$$

Total force exerted by all N molecules —

$$f = \Sigma \frac{mv_x^2}{\ell} = \frac{m}{\ell}\left[\frac{\Sigma v_x^2}{N}\right] \times N$$

$$f = \frac{mN}{\ell} v_{x\text{-rms}}^2$$

Avg. force exerted by all molecules, $f = \frac{mN}{\ell} v_{x\text{-}rms}^2$

$\Rightarrow f = \frac{mN}{\ell}\left[\frac{v_{rms}^2}{3}\right]$ $\quad \{ v_{rms}^2 = 3 v_{x\text{-}rms}^2 \}$

Avg. pressure applied by all molecules. $P = \frac{f}{A} = \frac{mN}{\ell A}\left[\frac{v_{rms}^2}{3}\right]$

m = mass of one molecule
M = mass of one mole
N = Total no. of molecules
n = Total no. of moles
v_{rms} = RMS speed
$v_{x\text{-}rms}$ = RMS speed along x-axis
P = Pressure Exerted by gas, ρ = Density of gas

$P = \frac{\text{mass}}{\text{Vol.}}\left[\frac{v_{rms}^2}{3}\right]$ $\quad \{ \ell \cdot A = \text{Vol. } (V),\ \text{mass} = mN = nN \}$

$P = \frac{1}{3}\rho v_{rms}^2$

$$P = \frac{1}{3}\rho v_{rms}^2$$

$$\{v_{rms} = \sqrt{\frac{3RT}{M}}\}$$

$$\Rightarrow P = \frac{1}{3}\rho \frac{3RT}{M}$$

$$P = \frac{\rho RT}{M}$$

$$P = \frac{n\rho RT}{nM}$$

$$P = \frac{nRT}{\left(\frac{mass}{\rho}\right)}$$

$$\{mass = nM = Nm\}$$

$$\Rightarrow P = \frac{nRT}{V}$$

$$\Rightarrow PV = nRT$$

$$P = \frac{1}{3}\rho v_{rms}^2 = \frac{\rho RT}{M} = \frac{nRT}{V}$$

→ Use when no. of moles, temp & vol. are either given or asked.

Use when either v_{rms} is given or asked

Use when density & temp are either given or asked

Kinetic Energy of Gas

↳ Translational k.E

Trans. k.E of one molecule, $k.E_0 = \frac{1}{2} m v^2$

Total translational k.E of all N molecules, $k.E = \sum \frac{1}{2} m v^2$

$$k.E = \frac{1}{2} m \left[\frac{\sum v^2}{N}\right] \times N$$

$$= \frac{1}{2} (mN) v_{rms}^2 \quad \left\{v_{rms} = \sqrt{\frac{\sum v^2}{N}}\right\}$$

$$k.E = \frac{1}{2} (\text{mass}) \cdot v_{rms}^2 = \frac{1}{2} nM v_{rms}^2$$

$$k.E = \frac{1}{2} n \cancel{M} \left(\frac{3RT}{\cancel{M}}\right)$$

$$k.E = \frac{1}{2} \text{mass} \, v_{rms}^2$$

$\{\text{mass} = mN = nM\}$

$k.E \propto T$

$k.E \propto n$

$$k.E = \frac{3}{2} nRT$$

Meaning of temperature

Total translational k.E of n moles, $k.E = \frac{3}{2} nRT$

Total Translational k.E of 1 mole, $k.E = \frac{3}{2} RT$

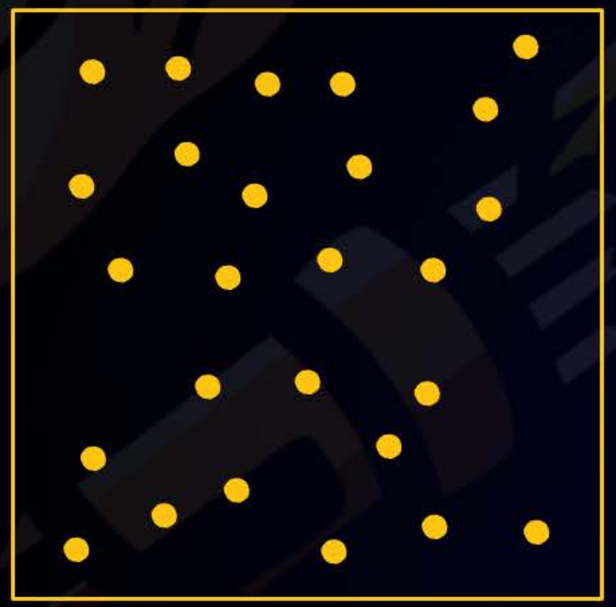
Total trans. k.E of N_A molecules, $k.E = \frac{3}{2} RT$

Avg trans. k.E of 1 molecule, $k.E = \frac{3}{2}\left(\frac{R}{N_A}\right)T$

Avg trans. k.E of one molecule, $k.E = \frac{3}{2} k_B T$

Temp. represents the avg k.E of molecules.

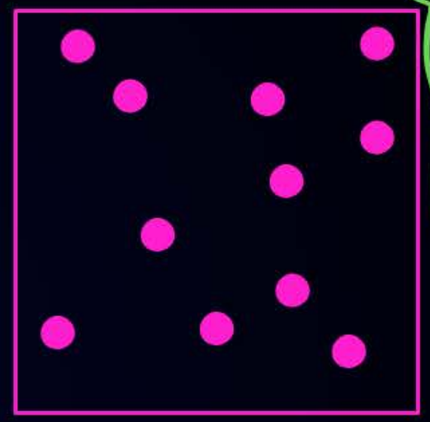
Avg trans. k.E per molecule $\propto T$.



T_A

$T_A < T_B$

Heat transfer



T_B

A

B

Total Energy = 100J

Energy per molecule = 1J

Total Energy = 10J

Energy per molecule = 2J.

Energy always transfer from higher temp. to lower temp.

Temp. total energy represent नहीं करता है। temp. molecules की avg. k.E represent करता है।

Gas Laws

Boyle's Law

T = Const.

n = Const.

At a given temperature, the pressure of a given mass of a gas is inversely proportional to its volume. This is known as Boyle's law.

If T = Const. –

$$P \propto \frac{1}{V}$$

$$P_1V_1 = P_2V_2$$

$$P = \frac{1}{3}\rho V_{rms}^2$$

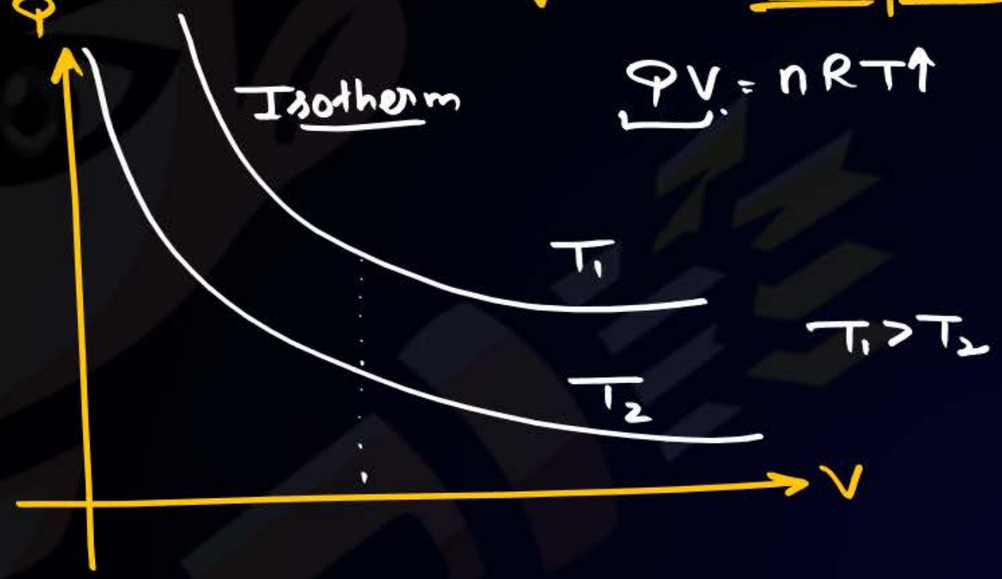
At const. temp. – V_{rms} = Const.

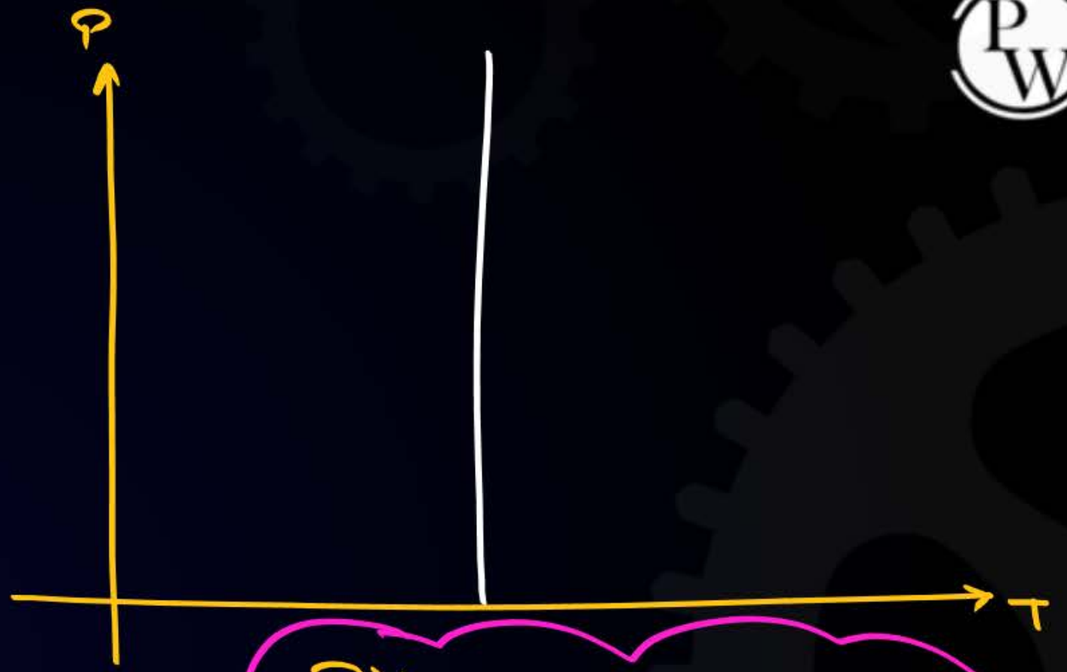
$$P \propto \rho \Rightarrow P \propto \frac{mass}{V}$$

$$P \propto \frac{1}{V}$$

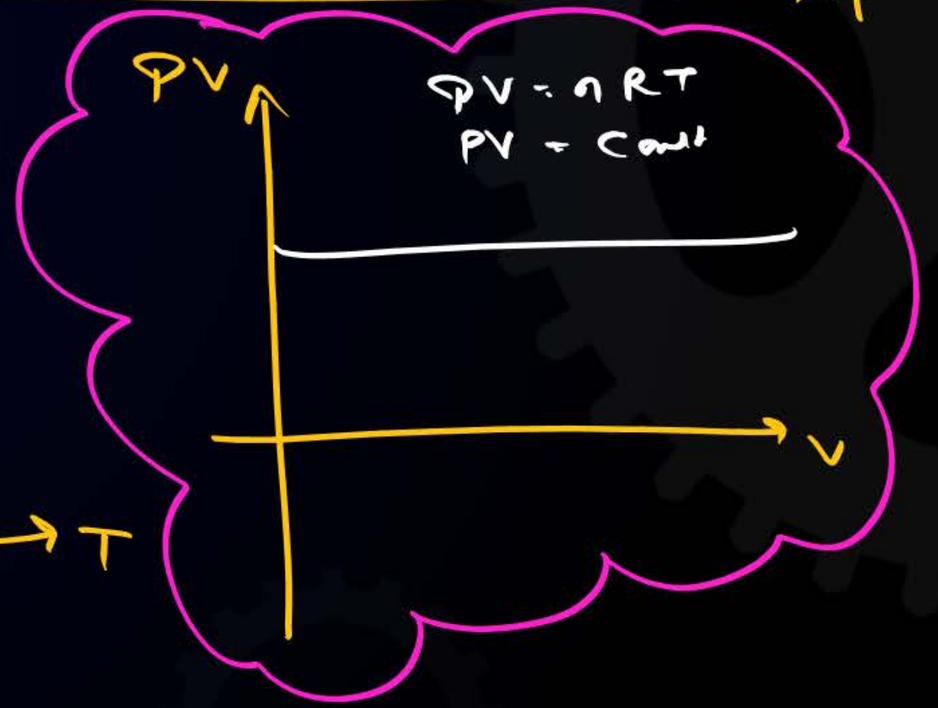
Graphs

$P \propto \frac{1}{V}$









Charle's Law

$P = \text{const}$ $n = \text{const}$

At a given pressure, the volume of a given mass of a gas is proportional to its absolute temperature. This is known as Charles's law.

$P = \text{const}$, $n = \text{const}$.

$$V \propto T$$

$$P = \frac{1}{3} \rho v_{rms}^2$$

$$\Rightarrow v_{rms}^2 \propto \frac{1}{\rho} \quad [P = \text{const}]$$

$$\Rightarrow \frac{3RT}{M} \propto \frac{1}{\left(\frac{\text{mass}}{V}\right)}$$

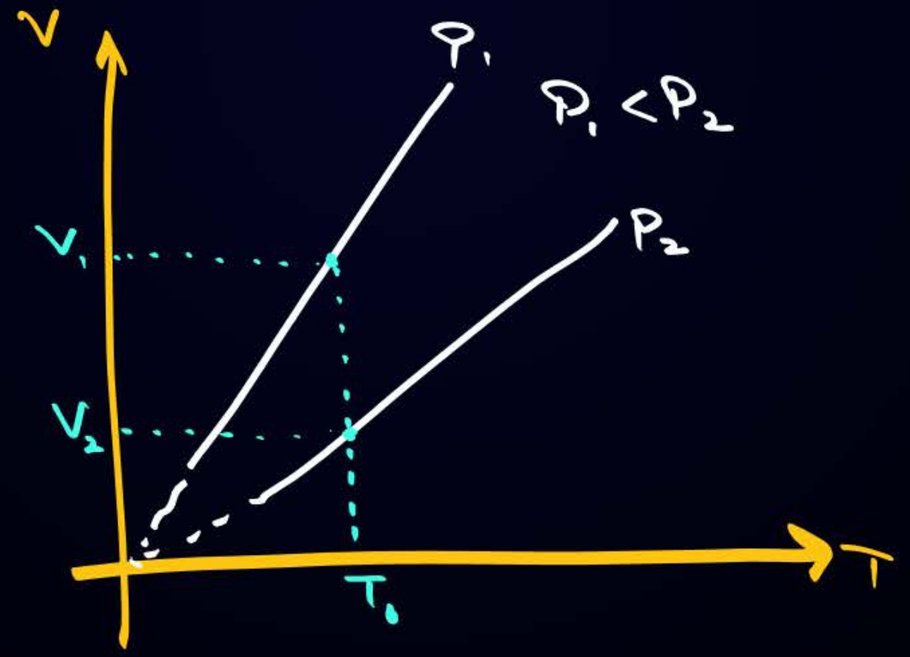
$$T \propto V$$

Graphs



$\downarrow P \uparrow V = nRT$





Charle's Law for Pressure

V = Const.

n = Const.

At a given volume, the pressure of a given mass of a gas is proportional to its absolute temperature. This is known as Charles's law for pressure.

V = Const., n = Const

$P \propto T$

$$P = \frac{1}{3} \rho V_{rms}^2$$

$$P = \frac{1}{3} \rho \left(\frac{3RT}{M} \right)$$

$P \propto T$

Vol. = Const

$\frac{\text{mass}}{\text{Vol.}} = \rho$ = Const

Avogadro's Law

At the same temperature and pressure, equal volumes of all gases contain equal number of molecules. This is known as Avogadro's law.

$P = \text{Const}, \quad T = \text{Const}$

$V \propto n$

$$P = \frac{1}{3} \rho V_{rms}^2$$

$$P = \frac{1}{3} \rho \cdot \frac{3RT}{M}$$

$$\rho = \text{Const}$$

$$\frac{\text{mass}}{V} = \text{Const.} \Rightarrow \frac{nM}{V} = \text{Const}$$

$n \propto V$

Graham's Law of Diffusion

When two gases at the same pressure and temperature are allowed to diffuse into each other, the rate of diffusion of each gas is inversely proportional to the square root of the density of the gas. This is known as Graham's law of diffusion.

$$\text{rate of diffusion} \propto v$$

$$\propto \frac{1}{\rho}$$

$$P = \frac{1}{3}\rho v_{rms}^2$$

$$v_{rms} = \sqrt{\frac{3P}{\rho}}$$

$$v_{rms} \propto \frac{1}{\rho}$$

Dalton's Law of Partial Pressure

Dalton's law of partial pressure says that the pressure exerted by a mixture of several gases equals the sum of the pressures exerted by each gas occupying the same volume as that of the mixture.

$$P = P_1 + P_2 + \dots P_n$$



Ideal Gas Equation

$PV = nRT$

Question

300k ⟶ 600k = 600-273 = 327°C

A perfect gas at 27°C is heated at constant pressure so as to double its volume. The final temperature of the gas will be, close to [2012]

A. 327°C

B. 200°C

C. 54°C

D. 300°C

Question

If 10^{22} gas molecules each of mass 10^{-26} kg collide with a surface (perpendicular to it) elastically per second over an area 1 m^2 with a speed 10^4 m/s, the pressure exerted by the gas molecules will be of the order of:

[8 April 2019 I]

A 10^4 N/m^2

B 10^8 N/m^2

C 10^3 N/m^2

D 10^{16} N/m^2

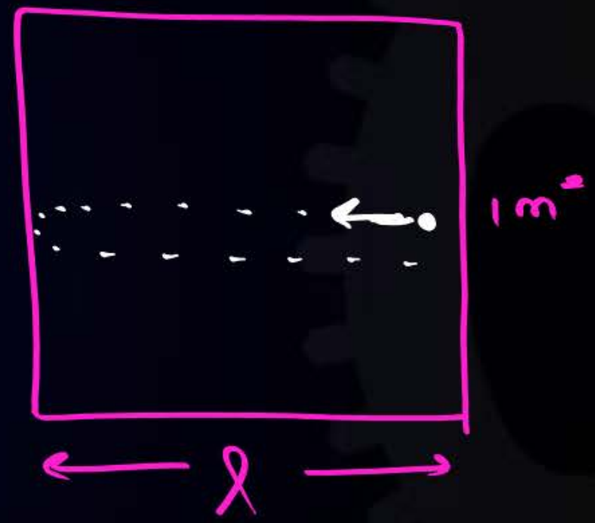
$P = \frac{1}{3}\rho v_{rms}^2$

$P = \frac{F}{A} = \frac{\Delta P}{A\,\Delta t}$

$= \frac{N \times mv}{A\,\Delta t}$

$= \frac{10^{22} \times 10^{-26} \times 10^4}{1 \times 1}$

$= 1$



Question

HW-1

Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to:

[9 Jan 2020 II]

A 3.67

B 1.83

C 2.3

D 4.67

Question

H.W-2

An ideal monatomic gas is confined in a cylinder by a spring-loaded piston of cross section 8.0×10^{-3} m^2. Initially the gas is at 300 K and occupies a volume of 2.4×10^{-3} m^{-3} and the spring is in its relaxed state (figure). The gas is heated by a small heater until the piston moves out slowly by 0.1 m. Calculate the final temperature of the gas. The force constant of the spring is 8000 N m^{-1}, and the atmospheric pressure is 1.0×10^5 N m^{-2}. The cylinder and the piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and the cylinder. Neglect any heat-loss through the lead wires of the heater. The heat capacity of the heater coil is negligible.



Question

H.W-3

Two glass bulbs of equal volume are connected by a narrow tube and are filled with a gas at 0°C and a pressure of 76 cm of mercury. One of the bulbs is then placed in melting ice and the other is placed in water bath maintained at 62°C. What is the new value of the pressure inside the bulbs? The volume of the connecting tube is negligible.

Question

H.W-4

Two closed containers of equal volume of air are initially at 1.05×10^5 Pa, and 300K temperature. If the containers are connected by a narrow tube and one container is maintained at 300 K temperature and other at 400 K temperature. Find the final pressure in the containers.

Question

d.w-5

A glass container encloses a gas at a pressure of 8×10^5 Pa and 300 K temperature. The container walls can bear a maximum pressure of 10^6 Pa. If the temperature of container is gradually increased, find the temperature at which container will break.

Question

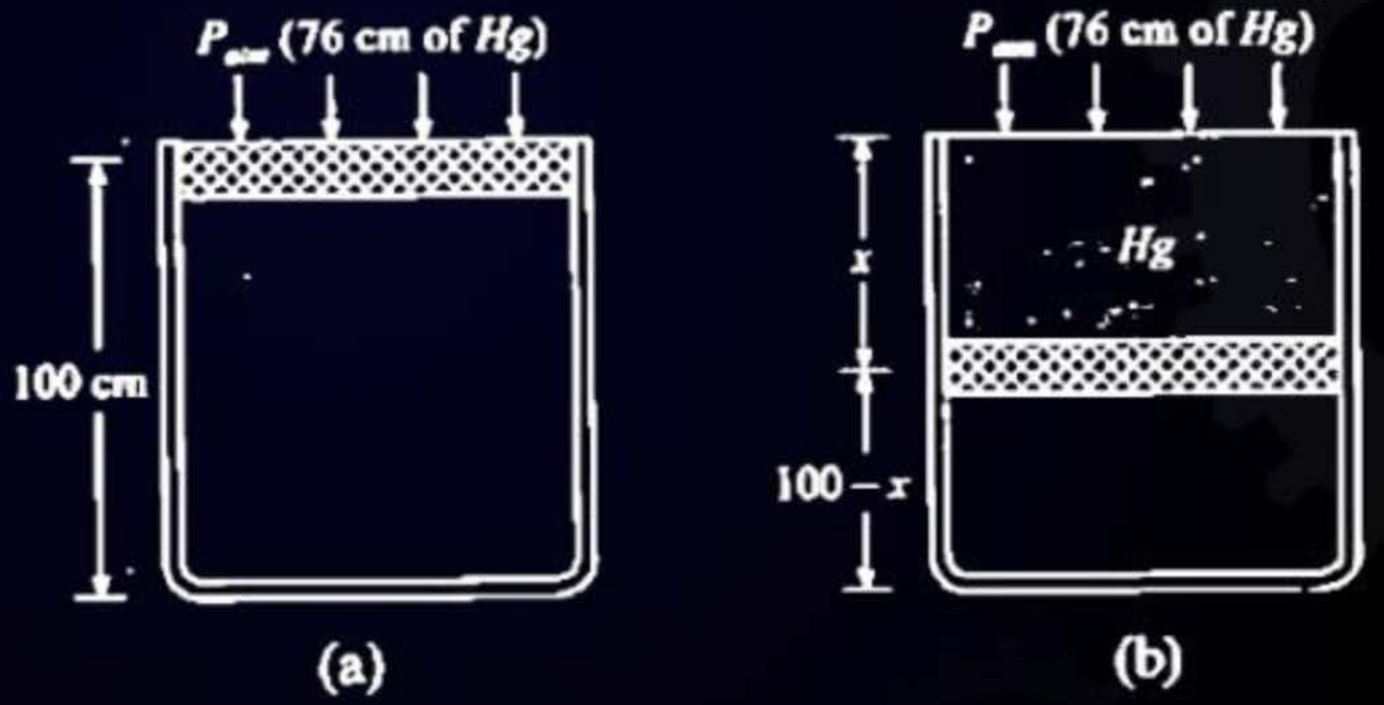
H.W - 6

Find the minimum attainable pressure of one mole of an ideal gas if during its expansion its temperature and volume are related as $T = T_0 + \alpha V^2$ where T_0 and α are positive constants.

Question

H.W-7

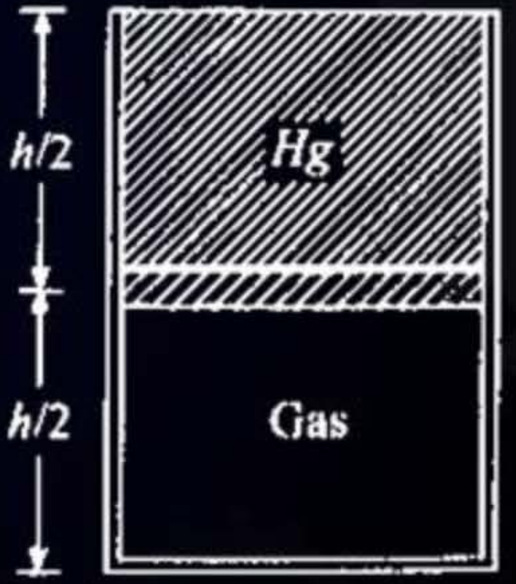
A vertical cylinder of height 100 cm contains air at a constant temperature and its top is closed by a frictionless piston at atmospheric pressure (76 cm of Hg) as shown in figure. If mercury is slowly poured on the piston, due to its weight air is compressed. Find the maximum height of the mercury column which can be put on the piston.



Question

H.W-8

A vertical hollow cylinder of height 1.52 m is fitted with a movable piston of negligible mass and thickness. The lower part of cylinder contains an ideal gas and the upper part is filled with mercury as shown in figure. Initially the temperature of system is 300 K and the lengths of gas and mercury columns are equal. Find the temperature to which system is raised so that half of mercury overflows. Take atmospheric pressure is 76 cm of Hg and neglect thermal expansion of mercury.



Question

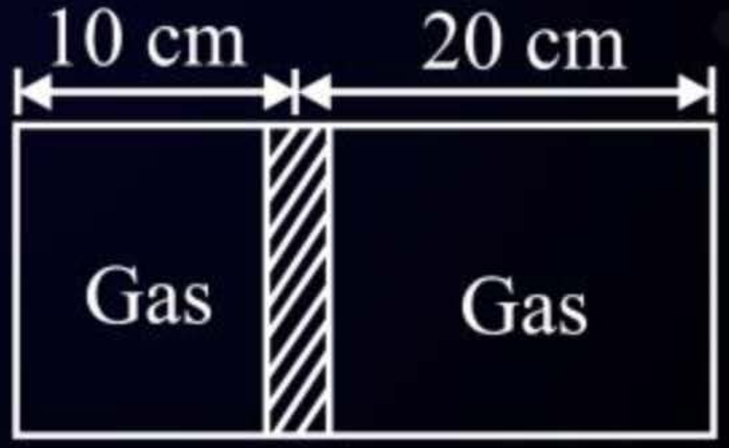
H.W-9

A uniform tube closed at one end, contains a pallet of mercury 10 cm long. When the tube is kept vertically with the closed end of tube upward, the length of air column trapped by mercury and the closed end of the tube is 20 cm. If the tube is inverted so that its open end becomes upward. Find the final length of the air column trapped. Take atmospheric pressure to be 76 cm of Hg.

Question

H.W-10

Figure shows a horizontal cylindrical container of length 30 cm, which is partitioned by a tight fitting separator. The separator is diathermic but conducts heat very slowly. Initially the separator is in the state shown in figure. The temperature of left part of cylinder is 100 K and that on right part is 400 K. Initially the separator is in equilibrium. As heat is conducted from right to left part, separator displaces to the right. Find the displacement of separator after a long two when gases on the parts of cylinder are in thermal equilibrium.



QUESTION (JEE Mains – Apr. 13, 2023 (II))

H.W-1

Two plates A and B have thermal conductivities 84 $Wm^{-1}K^{-1}$ and 126 $Wm^{-1}K^{-1}$ respectively. They have same surface area and same thickness. They are placed in contact along their surfaces. If the temperatures of the outer surfaces of A and B are kept at 100° C and 0°C respectively, then the temperature of the surface of contact in steady state is _____ °C.



$$R_1 + R_2 = \frac{l}{A}\left[\frac{1}{84} + \frac{1}{126}\right]$$

$$= \frac{l}{A}\left[\frac{3+2}{252}\right]$$

$$R_1 + R_2 = \frac{5l}{252A}$$

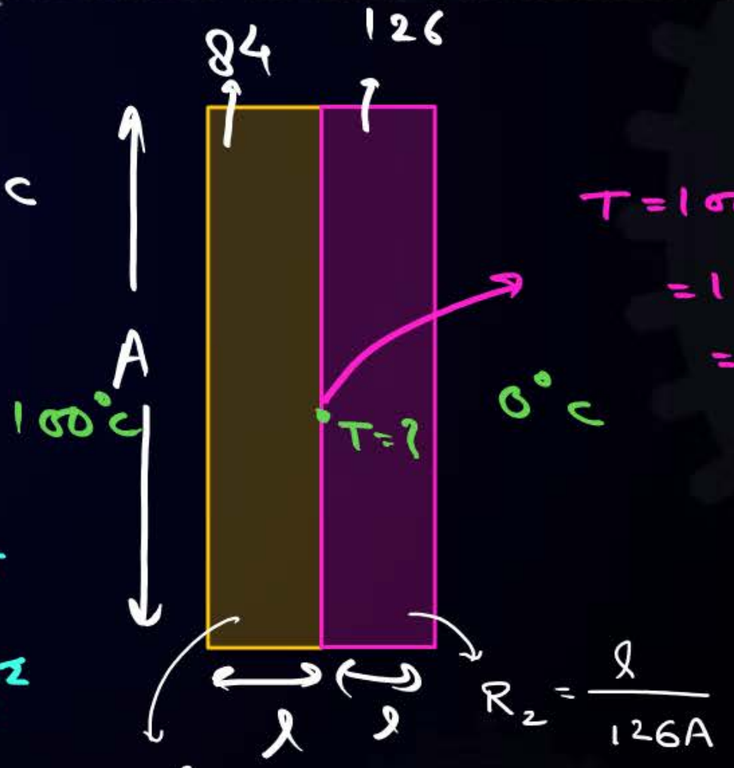
$R_1 + R_2 \longrightarrow 100°C$

$1 \longrightarrow \frac{100}{R_1 + R_2}$

$R_1 \longrightarrow \frac{100}{R_1 + R_2} * R_1 = \Delta T$

$\frac{100}{\left(\frac{5l}{252A}\right)} * \frac{l}{84A} = \frac{100 \times 252}{5 \times 84}$

$= 60°C$



$T = 100 - \Delta T$

$= 100 - 60$

$= 40°C$ Ans.

QUESTION (JEE Mains – March. 17, 2021 (I)

H.W-2

Two identical metal of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is:

A. $\frac{K_1+K_2}{K_1K_2}$

B. $\frac{K_1+K_2}{2K_1K_2}$

C. $\frac{2K_1K_2}{K_1+K_2}$

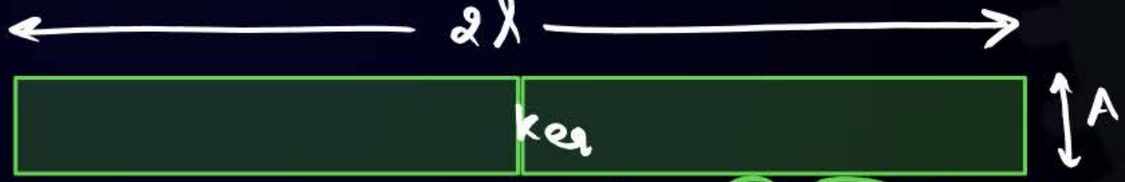
D. $\frac{K_1K_2}{K_1+K_2}$

$R_{eq} = R_1 + R_2$

$R_{eq} = \frac{\ell}{k_1 A} + \frac{\ell}{k_2 A}$ - ①



$R_{eq} = \frac{2\ell}{k_{eq} A}$ - ②



From ① & ② → $\frac{2\ell}{k_{eq} A} = \frac{\ell}{k_1 A} + \frac{\ell}{k_2 A}$ ⇒ $k_{eq} = \frac{2k_1k_2}{k_1+k_2}$

Ans

QUESTION (JEE Mains – June. 29, 2022 (I))

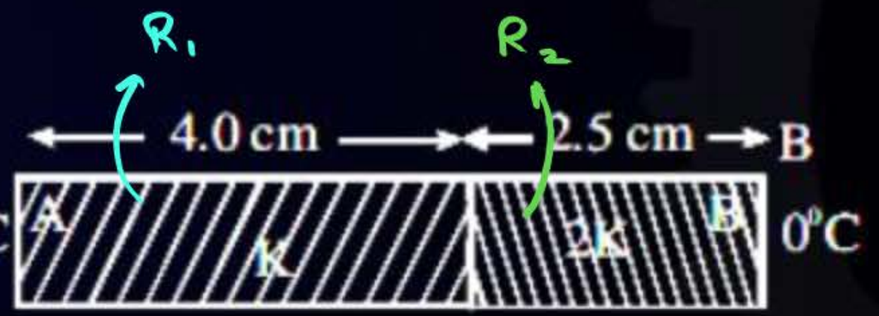
d.w-3

As per the given figure, two plates A and B of thermal conductivity K and 2K are joined together of form a compound plate. The thickness of plates are 4.0 cm and 2.5 cm respectively and the area of cross-section is 120 cm^2 for each plate. The equivalent thermal conductivity of the compound plate is $\left(1+\frac{5}{\alpha}\right)K$, then the value of α will be

21

$R_{eq} = R_1 + R_2$

$R_{eq} = \frac{4}{K \times 120} + \frac{2.5}{2K \times 120}$ - ①



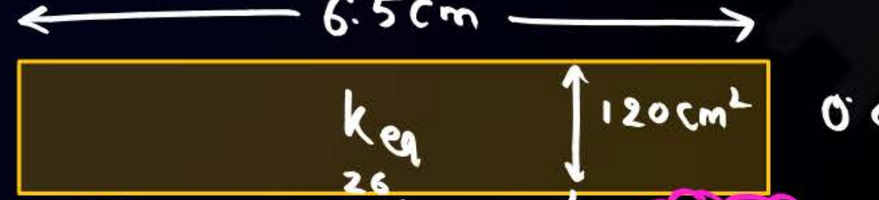
$R_{eq} = \frac{6.5}{k_{eq} \cdot 120}$ - ②

$\Rightarrow \frac{6.5}{k_{eq}} = \frac{10.5}{2K}$

$\Rightarrow k_{eq} = \frac{13}{10.5}$

From ① & ② →

$\frac{6.5}{k_{eq} \times 120} = \frac{4}{K \times 120} + \frac{2.5}{2K \times 120}$



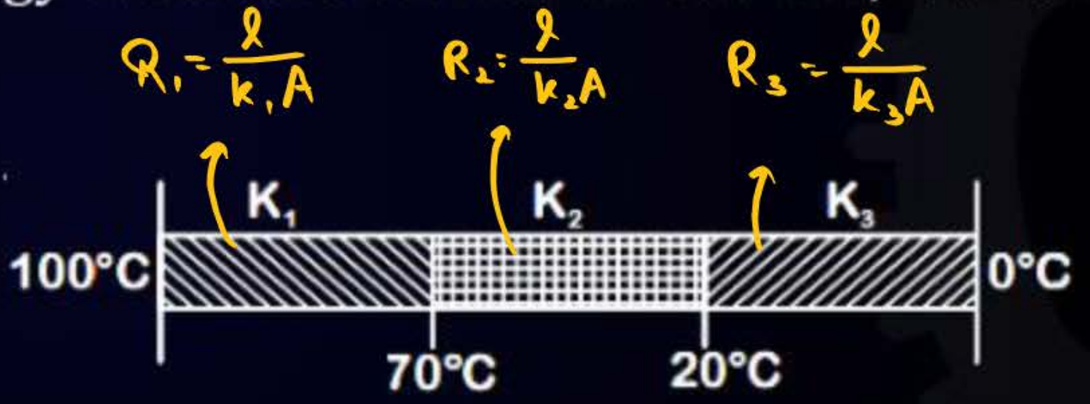
100°C

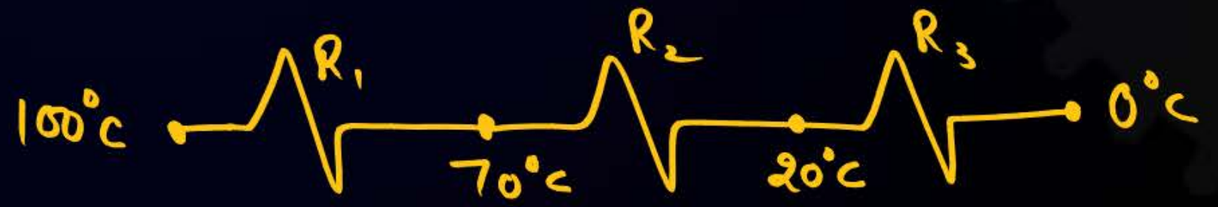
$1 + \frac{5}{\alpha} = \frac{13}{10.5} \Rightarrow \frac{5}{\alpha} = \frac{13}{10.5} - 1 = \frac{5}{21} \Rightarrow \alpha = 21$ Ans

QUESTION (JEE Mains – Sep. 06, 2020 (II))

H.W-4

Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1, K_2 and K_3 respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at 100°C and the other at 0°C (see figure). If the joints of the rod are at 70°C and 20°C in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1, K_2 and K_3 is



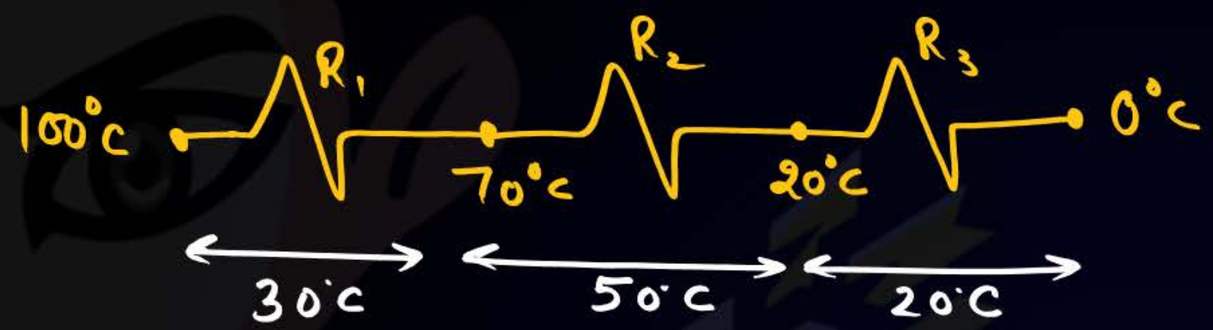


(A) $K_1 : K_3 = 2 : 3$; $K_2 : K_3 = 2 : 5$

(B) $K_1 < K_2 < K_3$

(C) $K_1 : K_2 = 5 : 2$; $K_1 : K_3 = 3 : 5$

(D) $K_1 > K_2 > K_3$



In series —

$$\Delta T \propto R$$

$$\Rightarrow R_1 : R_2 : R_3 = 30 : 50 : 20$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{3}{5}$$

$$\frac{\frac{\ell}{k_1 A}}{\frac{\ell}{k_2 A}} = \frac{3}{5} \Rightarrow \frac{k_2}{k_1} = \frac{3}{5} \Rightarrow \boxed{\frac{k_1}{k_2} = \frac{5}{3}}$$

$$\frac{R_2}{R_3} = \frac{5}{2}$$

$$\frac{\frac{\ell}{k_2 A}}{\frac{\ell}{k_3 A}} = \frac{5}{2} \Rightarrow \frac{k_3}{k_2} = \frac{5}{2} \Rightarrow \boxed{\frac{k_2}{k_3} = \frac{2}{5}}$$

$$\frac{R_1}{R_3} = \frac{3}{2} \Rightarrow \frac{\cancel{\ell}/k_1 \cancel{A}}{\cancel{\ell}/k_3 \cancel{A}} = \frac{3}{2} \Rightarrow \frac{k_3}{k_1} = \frac{3}{2}$$

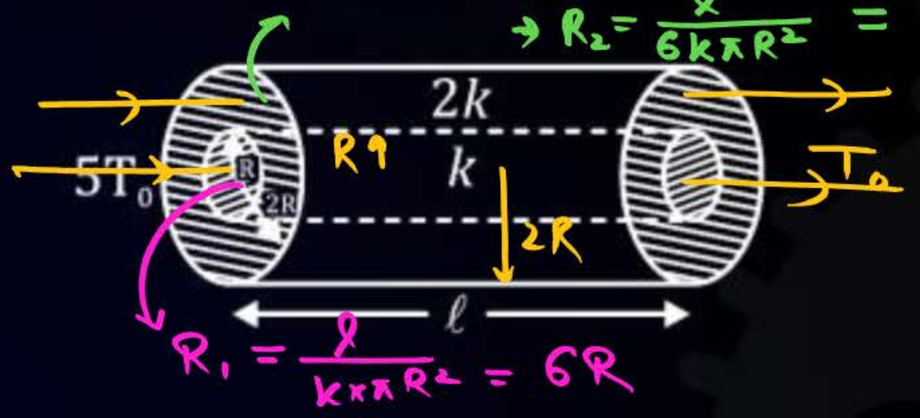
$$\Rightarrow \boxed{\frac{k_1}{k_3} = \frac{2}{3}}$$

Question ★★ Y1.W-5

Find.

(1) Rate of heat flow

(2) Equivalent thermal conductivity.

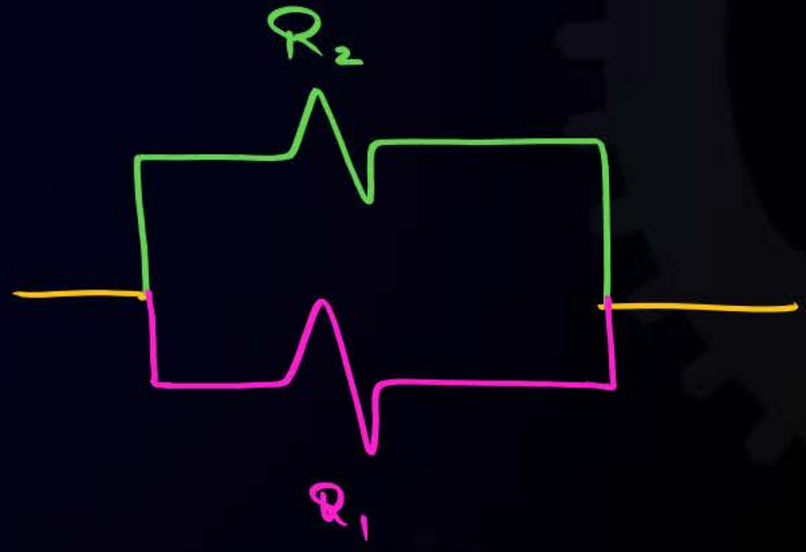


1) $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{\left[\frac{\ell}{k\times\pi R^2}\right]} + \frac{1}{\left[\frac{\ell}{6k\pi R^2}\right]} = \frac{k\pi R^2}{\ell} + \frac{6k\pi R^2}{\ell}$

$\Rightarrow \frac{1}{R_{eq}} = \frac{7k\pi R^2}{\ell} \Rightarrow R_{eq} = \frac{\ell}{7k\pi R^2}$

$\frac{dQ}{dt} = \frac{\Delta T}{R_{eq}} = \frac{5T_0 - T_0}{\left(\frac{\ell}{7k\pi R^2}\right)} = \frac{28k\pi R^2 T_0}{\ell}$ Ans

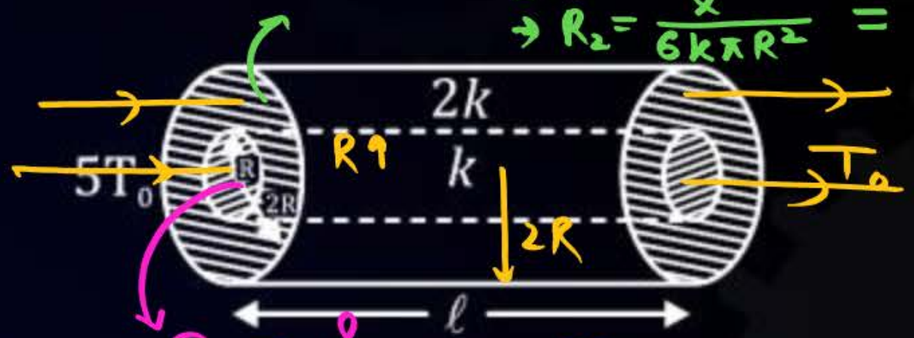


Question ★★ Y1.W-5

Find.

(1) Rate of heat flow

(2) Equivalent thermal conductivity.



$R_2 = \frac{\ell}{2k[\pi(2R)^2 - \pi R^2]}$

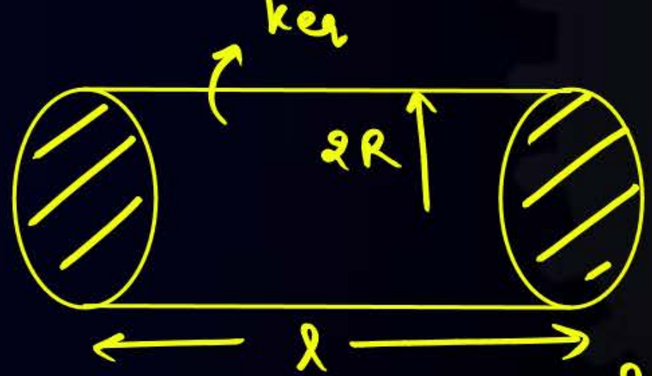
$\Rightarrow R_2 = \frac{\ell}{6k\pi R^2} = R$

$R_1 = \frac{\ell}{k\pi R^2} = 6R$

2) $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{\left[\frac{\ell}{k\pi R^2}\right]} + \frac{1}{\left[\frac{\ell}{6k\pi R^2}\right]} = \frac{k\pi R^2}{\ell} + \frac{6k\pi R^2}{\ell}$

$\Rightarrow \frac{1}{R_{eq}} = \frac{7k\pi R^2}{\ell} \Rightarrow R_{eq} = \frac{\ell}{7k\pi R^2}$ - ①



$R_{eq} = \frac{\ell}{k_{eq}\cdot\pi(2R)^2} = \frac{\ell}{4k_{eq}\pi R^2}$ - ②

From ① & ② - $\frac{\ell}{4k_{eq}\pi R^2} = \frac{\ell}{7k\pi R^2} \Rightarrow k_{eq} = \frac{7k}{4}$ Ans

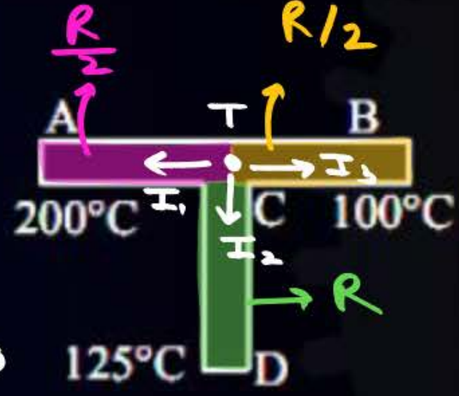
QUESTION (JEE Mains – Aug. 27, 2021 (I))

H.W-6

R ... K.W^{-1}

A rod CD of thermal resistance 10.0 ~~K.W^{-1}~~ is joined at the middle of an identical rod AB as shown in figure, The end A, B and D are maintained at 200°C, 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is

Applying KCL at junction — $I_1 + I_2 + I_3 = 0$



Now,

$$I_1 = \frac{\Delta T}{R_1} = \frac{T-200}{(R/2)} = \frac{2(T-200)}{R}$$

$$I_2 = \frac{\Delta T}{R_2} = \frac{T-125}{R}$$

$$I_3 = \frac{\Delta T}{R_3} = \frac{T-100}{(R/2)} = \frac{2(T-100)}{R}$$

$$\frac{2(T-200)}{R} + \frac{T-125}{R} + \frac{2(T-100)}{R} = 0$$

$$\Rightarrow 2T - 400 + T - 125 + 2T - 200 = 0$$

$$\Rightarrow 5T = 725$$

$$\Rightarrow T = 145^\circ C$$

$$I_2 = P = \frac{T-125}{R} = \frac{145-125}{10}$$

$$= \frac{20}{10} = 2 \text{ watts}$$

Ans

Question * Challenged

H.W-7

The ends Q and R of two thin wires, PQ and RS, are soldered (joined) together. Initially each of the wires has a length of 1 m at 10°C. Now the end P is maintained at 10°C, while the end S is heated and maintained at 400°C. The system is thermally insulated from its surroundings. If the thermal conductivity of wire PQ is twice that of the wire RS and the coefficient of linear thermal expansion of PQ is $1.2 \times 10^{-5}\ K^{-1}$ (α), the change in length of the wire PQ is : **[Adv. 2016]**

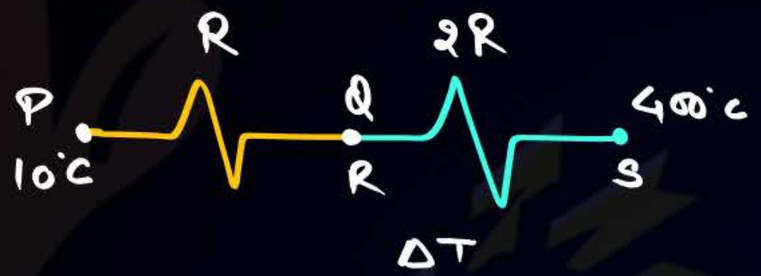
A 0.78 mm

B 0.90 mm

C 1.56 mm

D 2.34 mm

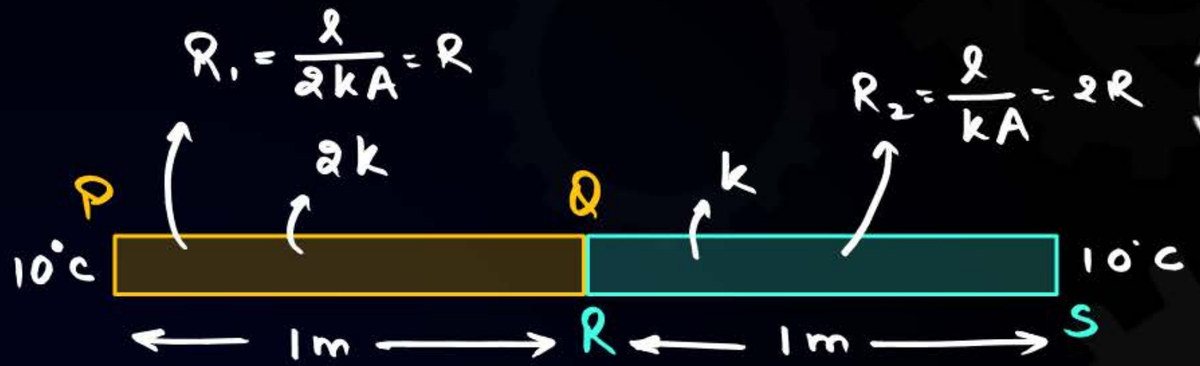
Ans : (A)

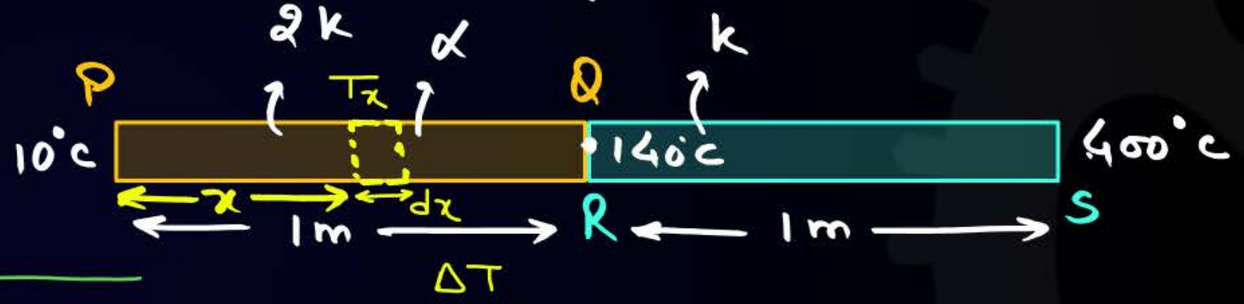


$3R \longrightarrow 390^\circ c$

$R \longrightarrow \frac{390}{3} = 130^\circ c$

$T_Q = 10 + 130 = 140^\circ c$





$1m \longrightarrow 130^\circ c$

$x \longrightarrow 130x$

Change in length of small element dx →

$\delta(dx) = \alpha\, dx\, \Delta T$

$\int \delta(dx) = \int_0^1 \alpha\, dx\, (130x)$

$\Delta x = \alpha \times 130 \times \left[\frac{x^2}{2}\right]_0^1 = 1.2 \times 10^{-5} \times 130 \times \frac{1}{2}$

$= 78 \times 10^{-5}\ m$ Ans

QUESTION (JEE Mains – Sep. 03, 2020 (II))

A metallic sphere cools from 50°C to 40°C in 300 s. If atmospheric temperature around is 20°C, then the sphere's temperature after the next 5 minutes will be close to :

(A) 31°C

(B) 33°C

(C) 28°C

(D) 35°C

5 minutes = 300 s.

$50^\circ C \longrightarrow 40^\circ C : 300\ s$

$$-\frac{dT}{dt} = k(T - T_0) \Rightarrow \frac{50-40}{300} = k\left[\left(\frac{50+40}{2}\right) - 20\right]$$

$40^\circ C \longrightarrow T : 300\ s$

$$-\frac{dT}{dt} = k(T - T_0) \Rightarrow \frac{40-T}{300} = k\left[\left(\frac{40+T}{2}\right) - 20\right]$$

$$\Rightarrow \frac{10}{40-T} = \frac{25}{\left(\frac{40+T-40}{2}\right)} \Rightarrow \frac{10}{40-T} = \frac{50}{T}$$

$10T = 50(40-T)$

$10T = 2000 - 50T$

$60T = 2000$

$T = \frac{200}{6} = \frac{100}{3}$

QUESTION (JEE Mains – July. 27, 2021 (I))

A body takes 4 min. to cool from 61°C to 59°C. If the temperature of the surroundings is 30°C, the time taken by the body to cool from 51°C to 49°C is:

A. 6 min

B. 3 min

C. 4 min

D. 8 min

$61^\circ C \longrightarrow 59^\circ C : 4\,min \Rightarrow \frac{61-59}{4} = k\left[\left(\frac{61+59}{2}\right)-30\right]$

$51^\circ C \longrightarrow 49^\circ C : t \Rightarrow \frac{51-49}{t} = k\left[\left(\frac{51+49}{2}\right)-30\right]$

$\Rightarrow \frac{t}{4} = \frac{3\beta}{2\beta} \Rightarrow t = 6\,min.$ Ans

QUESTION (JEE Mains – 2014)

Hot water cools from 60°C to 50°C in the first 10 minutes and to 42°C in the next 10 minutes. The temperature of the surroundings is:

(A) 25°C

(B) 10°C ✓

(C) 15°C

(D) 20°C

$60^\circ C \longrightarrow 50^\circ C : 10\ min. \Rightarrow \frac{60-50}{10} = K\left[\left(\frac{60+50}{2}\right) - T_0\right]$

$50^\circ C \longrightarrow 42^\circ C : 10\ min. \Rightarrow \frac{50-42}{10} = K\left[\left(\frac{50+42}{2}\right) - T_0\right]$

$\Rightarrow \frac{10}{8} = \frac{55-T_0}{46-T_0}$

$\Rightarrow 460 - 10T_0 = 440 - 8T_0$

$\Rightarrow 2T_0 = 20$

$T_0 = 10^\circ C$

Homework

DPP:	~~NO~~ DPP - 1
MODULE:	Th. Prob. - Complete Exercise - 1, 2, 3.

Revision Test No.23:	**COM**
(5 Jan 2024, Friday)	**Lectures – 1, 2, 3, 4, 5, 6, 7**


THANK YOU